

2. The forward price is a perfect predictor of the future mean—

This assumption implies that options will be priced with probabilities centered at the corresponding forward price. Sometimes, however, when the forward price is well removed from the current price, it may not be reasonable to assume that a price change equal to the difference between the forward and spot price is the most likely market outcome. Frequently, there may be good reason to assume that some price between the forward and spot price is more likely than the forward price. If this is true, out-of-the-money options (puts if the forward price is higher and calls if it is lower) may be underpriced. The Brazilian interest rate trade Mai described provided a good example of such an opportunity.

3. Volatility scales with the square root of time—This assumption, which is embedded in option pricing, may be reasonable for shorter time intervals. For longer time periods, however, this volatility assumption may understate potential volatility, particularly if current volatility is low, for two reasons. First, the longer the time period, the more likely volatility will revert to the mean from current low levels. Second, longer periods allow for more opportunity for trends to result in larger price moves than implied by the volatility assumption.

4. The trend can be ignored in the volatility calculation—Option pricing models gauge the probability of price moves of a given magnitude based on volatility and time. Trend is not part of the calculation. The implicit assumption is that the direction of daily price changes is random. Consequently, a trending market can result in price moves that would be deemed improbable by the pricing model. If there is a reason to expect a trend, then out-of-the-money options are likely to be underpriced. Mai's Canadian dollar option trade was based on this premise.

5. Current correlations are good predictors of future correlations—

Some market correlations tend to be fairly consistent (e.g., gold and platinum tend to be positively correlated over the broad spectrum of time), while other market pairs may exhibit variable correlation patterns (e.g., the Australian dollar and Swiss franc that were part of Mai's *worst of* basket trade). Correlation-based trades will tend to assume